

science

Swimming for Cancer

The tadpole test.

By Paul Boutin

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When I step into the Molecular Sciences Institute in downtown Berkeley, I find a bunch of mixed-breed postdocs—math, physics, engineering, computers—in street duds. The genomics research lab looks like a high-school classroom. There are benches covered with pipette stands, a few laptops, thousands of pinky-sized plastic test tubes, a few Bunsen burners, and a deep freezer for cellular cultures. The biologist in the room turns out to be the guy who looks like (and is) a motorcycle mechanic, casually whipping up a batch of custom DNA in what looks like (and is) a green plastic ice bucket. The bucket holds what could become an early-warning blood test for cancer.

MSI's [research](#) seeks to discover and model the molecular processes inside living cells. The nonprofit lab has about 25 researchers—and at least \$20 million in funding. But instead of using clean rooms and supercomputers, the MSI team creates custom life forms with buckets and outdated Macs. Welcome to the new biotech—part IT project, part cooking class.

The only mind-bending piece of equipment in sight is a DNA synthesizer, a yellowing plastic appliance (you can buy a [used](#) one for \$2,500 on eBay) that spits out strands of DNA from computer-generated sequences. It's like an inkjet printer for genes. You plug in bottles of the four basic [nucleotides](#)—A, G, C, T—as if they were toner cartridges, send a DNA sequence from your computer, and out comes the components of a custom double helix. (If that's too much work, even, you can order your custom DNA [online](#) and FedEx will deliver it two days later.)

MSI was founded in 1996 by acclaimed molecular biologist and future Nobel laureate Sydney Brenner, who believed in recruiting researchers from outside the field of biology. A few years back, one of Brenner's hires, a physicist by training, goaded his biologist labmates over their crude measuring tools. Why can't you guys identify any molecule as reliably as you can a DNA sequence? he asked. The question spurred the development of MSI's "tadpoles." These custom molecular widgets can detect trace molecular counts of substances—proteins, poisons, or inorganic stuff like lead—in concentrations too low for other current measurements to register.

A tadpole (technically known as a [protein-DNA chimera](#)) is a hybrid of two molecules. Its head is a protein designed to bind to one specific type of molecule. Its tail is a strip of DNA that serves as a chemical bar code. Despite its name, the tadpole isn't alive. It's a chemical sticky. Mix some tadpoles into a blood sample and their heads will stick to, say, the specific kind of protein that breaks loose into your blood as a prostate tumor develops—months before your doctor would notice anything funny down there. In the past, biologists would have struggled to find and count the protein heads. But the tadpoles' DNA tails stand out like price tags. "No other biological molecule can be quantified as easily, or with as much sensitivity, as DNA," Ian Burbulis, the biker biologist, explained to me.

Burbulis showed me how he makes the tadpoles. The easy part is printing out their tails on the DNA synthesizer. Burbulis grows the protein heads by genetically reprogramming bacteria. Doping a test tube full of them with another custom DNA sequence forces the bacteria to make the protein heads inside

their single-cell bodies. Burbulis fills a few test tubes with a broth of modified bacteria and nutrients and lets them fatten up for a day or so. Then he slaughters the herd, zapping them with a titanium probe that emits ultrasound waves. The probe ruptures the bacteria's cell walls. The dying bacteria spew protein heads into the test tube. Burbulis quickly gathers the heads and stores them on ice in the bucket. "It's just like cooking," he says. "The ingredients will spoil and you want the product to be fresh." When he's collected enough heads, he mixes them with the tails and then bakes the mix gently at 30 degrees Celsius for 12 to 16 hours. If he gets it right, the result is a nice warm batch of tadpoles.

To test your blood for cancer, a medical lab would mix the tadpoles with a single drop of your blood. A minute or two later the tester would wash away any tadpoles that hadn't bound to a target. To measure the remaining tadpoles—the ones that have latched onto cancer indicators, the tester would place the tadpole-bearing blood sample into a PCR ([polymerase chain reaction](#)) machine, a sort of incubator that replicates short DNA strands.

This is the genius part. Even if there were only a dozen tadpoles in your blood sample, the PCR would multiply their tails until there were enough (say a thousand or so) to be detected by standard lab gear. By dividing what he or she had just multiplied, the tester would know roughly how many tadpole tails—and hence how many cancer-indicating molecules—were in your blood sample. The whole process takes an afternoon at most. MSI refused to let me quote a number until rigorous trials are done, but I'd wager that tadpoles could be at least 10 times more sensitive than current lab tests at spotting cancer.

The challenges to bringing MSI's cancer test to market include mass-producing tadpoles on the cheap and surviving federal testing and approval. If the lab makes it past those hurdles, then its tadpoles could one day help doctors diagnose gastric, testicular, ovarian, breast, colon, rectal, lung, pancreatic, or other cancers much sooner. They could find trace-level warnings of other health problems, too, all as part of a standard blood test. Not bad for starting out in a green plastic bucket.

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